■ CHVN Paper Reviewer

Multi-Agent Scientific Paper Review

qwen3:8b · May 20, 2026 15:56

File	REVISTA_Project.pdf
Words	6,252
Sections Detected	preamble, introduction, methodology, experiments, results, discussion, conclusions, references, abstract,
Review Mode	DEEP
Model	qwen3:8b
Target Publisher	IEEE

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Editorial Decision

MAJOR REVISION

Weighted Score: 2.9 / 5.0
Confidence: 85%

Executive Summary

The manuscript presents a solar-powered refrigeration system for rural food preservation, with a claimed coefficient of performance (COP) of 4.72. However, the work lacks sufficient technical rigor, experimental validation, and reproducibility.

Decision Rationale

The manuscript requires major revisions to address critical issues in reproducibility, statistical validation, and ethical reporting. While the paper presents a novel system design, the lack of rigorous methodology, statistical analysis, and transparency in reporting significantly impacts its scientific validity and compliance with IEEE standards.

Evaluation Score Table

Scale: 0 = Not evaluable | 1 = Very weak | 2 = Weak | 3 = Acceptable | 4 = Good | 5 = Excellent

Criterion	Score	Rating	Visual
Originality & Novelty	**4.2**	Good	■■■■■
Manuscript Structure	**2.0**	Weak	■■■■■
Methodology & Equations	**3.5**	Good	■■■■■
Statistical Analysis	**1.5**	Very Weak	■■■■■
Results & Evidence	**3.5**	Good	■■■■■

Discussion & Conclusions	**2.5**	Weak	■■■■■
Figures & Tables	**4.2**	Good	■■■■■
Scientific Writing	**2.5**	Weak	■■■■■
References & Citations	**2.5**	Weak	■■■■■
Ethics & Transparency	**1.0**	Very Weak	■■■■■

Detailed Agent Reviews

TitleAbstractKeywordsReviewer

Title, Abstract, Keywords

Score: **4.2 / 5.0** | Confidence: 95%

Strengths:

- [TITLE] Clearly states the purpose and scope of the work, including the use of solar energy and the target application (rural food preservation).
- [ABSTRACT] Follows a structured format with clear background, objective, methods, results, and conclusions.
- [ABSTRACT] Quantifies results (e.g., 40% food loss, 4.72 COP, 286 L volume).
- [KEYWORDS] Includes domain-specific terms such as 'COP', 'R290', 'monitoreo IoT', and 'sistema fotovoltaico', enhancing discoverability.

Weaknesses:

- [TITLE] Could be more concise, as it exceeds the 10-20 word guideline and includes a redundant English translation.
- [ABSTRACT] Lacks a clear statement of novelty or comparison to existing solutions, which is critical for IEEE standards.
- [ABSTRACT] Does not mention the specific performance metrics of the IoT monitoring system or its integration with the refrigeration system.

Major Comments:

- [ABSTRACT] The claim of a 4.72 COP is significant, but the abstract does not specify the operating conditions or the basis for the thermodynamic analysis. This is critical for reproducibility and validation.
- [ABSTRACT] The phrase 'real coefficient of performance (COP) of 4.72' is not clearly defined in the abstract. The reader should be informed of the methodology used to calculate this value.

Minor Comments:

- [TITLE] The title includes a redundant English translation. It should be either in Spanish or English, not both.
- [KEYWORDS] The keyword 'Colombia' is specific to the study's location but may limit the global applicability of the work. Consider including broader terms such as 'off-grid communities' or 'solar-powered refrigeration'.

Recommendations:

- [TITLE] Revise the title to remove the redundant English translation and shorten it to meet the 10-20 word guideline.
- [ABSTRACT] Clarify the thermodynamic analysis and operating conditions in the abstract to support the COP claim.
- [ABSTRACT] Include a statement of novelty or comparison to existing solar-powered refrigeration systems in the abstract to meet IEEE standards.
- [ABSTRACT] Define the 'real coefficient of performance (COP)' in the abstract and specify the methodology used to calculate it.

StructureReviewer

Manuscript structure and logical coherence

Score: **2.0 / 5.0** | Confidence: 95%

Strengths:

- Title and abstract clearly state the problem of food preservation in rural areas using solar energy.

Weaknesses:

- Introduction is incomplete and lacks a clear research problem statement, hypothesis, and objectives.
- Methodology section is cut off and lacks detailed description of the six phases, making reproducibility questionable.
- Experiments section is fragmented and lacks a clear experimental design or procedure.

- Results section is incomplete and lacks quantitative data or analysis to support claims.
- Discussion section contains an equation (COPCarnot) without proper definition, units, or context, and lacks a coherent analysis of results.
- Conclusions are not supported by the results and lack specific contributions or implications.
- References are incomplete and contain placeholders or unformatted citations.
- Keywords are missing and not aligned with the content.
- Abstract is incomplete and lacks a clear statement of the research question or hypothesis.
- Title is cut off and does not fully reflect the content of the manuscript.

Major Comments:

- The manuscript lacks a clear hypothesis or research question, which is essential for a scientific paper.
- The methodology is not fully described, making it impossible to assess the validity or reproducibility of the work.
- The results section is incomplete and lacks quantitative data to support the claims made in the abstract and conclusions.
- The discussion section contains an equation without proper definition, units, or context, which undermines the technical rigor.
- The conclusions are not supported by the results and do not reflect the actual findings of the study.
- The references are incomplete and contain placeholders, which is unprofessional and affects the credibility of the work.
- The abstract is incomplete and does not clearly state the research question or hypothesis.

Minor Comments:

- The title is cut off and does not fully reflect the content of the manuscript.
- Keywords are missing and should be included to improve searchability and alignment with the content.

Recommendations:

- Complete the introduction to clearly state the research problem, hypothesis, and objectives.
- Provide a detailed description of the methodology, including the six phases, to ensure reproducibility.
- Include a complete experimental design and procedure in the experiments section.
- Add quantitative data and analysis in the results section to support the claims made in the abstract and conclusions.
- Define all equations, including units and context, in the discussion section.
- Ensure that the conclusions are directly supported by the results and reflect the actual findings of the study.
- Complete the references section with properly formatted citations.
- Include keywords that accurately reflect the content of the manuscript.

MethodologyReviewer

Methodology, experimental design, equations — holistic review

Score: **3.5 / 5.0** | Confidence: 90%

Strengths:

- The paper provides a clear and detailed description of the prototype design and construction process.
- The thermodynamic analysis and COP calculation are well-documented and align with the stated objectives.
- The IoT-based monitoring system is described with sufficient detail for potential replication.
- The paper includes a comprehensive comparison with existing technologies and highlights the advantages of the proposed solution.

Weaknesses:

- The 'Experiments' section is incomplete and lacks detailed experimental setup and procedure.
- The 'Results' section is brief and lacks detailed statistical analysis or confidence intervals.
- The paper does not provide a clear description of the experimental conditions or the number of trials conducted.

- The 'Discussion' section is somewhat repetitive and could benefit from more in-depth analysis of the results.
- The paper lacks a detailed description of the experimental setup, including the number of sensors, sampling frequency, and data collection methods.

Major Comments:

- The 'Experiments' section is incomplete and lacks detailed experimental setup and procedure. This is a critical gap in the methodology and limits the reproducibility of the work.
- The 'Results' section is brief and lacks detailed statistical analysis or confidence intervals. This weakens the validity of the findings.
- The paper does not provide a clear description of the experimental conditions or the number of trials conducted. This is essential for assessing the reliability of the results.
- The 'Discussion' section is somewhat repetitive and could benefit from more in-depth analysis of the results.
- The paper lacks a detailed description of the experimental setup, including the number of sensors, sampling frequency, and data collection methods. This is essential for reproducibility.

Minor Comments:

- Some equations are not consistently defined in the text, with some equations referenced without prior explanation.
- The paper could benefit from more detailed descriptions of the experimental conditions and the number of trials conducted.
- The 'Discussion' section could be more comprehensive in its analysis of the results.

Recommendations:

- Expand the 'Experiments' section to include detailed experimental setup, procedure, and conditions.
- Provide more detailed statistical analysis in the 'Results' section, including confidence intervals and effect sizes.
- Include a clear description of the experimental conditions and the number of trials conducted.
- Enhance the 'Discussion' section with more in-depth analysis of the results and their implications.
- Provide a more detailed description of the experimental setup, including the number of sensors, sampling frequency, and data collection methods.

StatisticsReviewer

Statistical methods, tests, and reporting quality

Score: **1.5 / 5.0** | Confidence: 80%

Strengths:

- Clear and detailed description of the system design and operation.
- Use of appropriate metrics such as COP, efficiency, and CV for performance evaluation.
- Comprehensive discussion of the environmental and practical implications of the design.
- Use of IoT for real-time monitoring and data collection.
- Comparison with existing literature and prior work (e.g., Zavaleta, Martínez Solís).

Weaknesses:

- No statistical tests explicitly reported in the manuscript, despite the presence of comparative claims.
- No confidence intervals reported for key metrics such as COP, efficiency, or power consumption.
- No justification for sample size (e.g., 30 independent runs) or power analysis.
- No mention of multiple comparison corrections (e.g., Bonferroni, Holm) for any statistical tests.
- No verification of test assumptions (e.g., normality, independence, homoscedasticity) for any statistical tests.
- No reproducibility details (e.g., seeds, runs, variance) for any simulations or experiments.
- No bootstrap methods or nonparametric approaches for uncertainty quantification.

Major Comments:

- The manuscript lacks any explicit statistical tests, despite the presence of comparative claims. This is a major issue for scientific validity and reproducibility.

- No confidence intervals are reported for key metrics such as COP, efficiency, or power consumption, which is essential for interpreting the precision of results.
- The sample size (e.g., 30 independent runs) is not justified or analyzed for power, which is a critical requirement for experimental design.
- No multiple comparison corrections are applied, which is a major concern for the validity of any statistical inferences.
- No verification of test assumptions (e.g., normality, independence, homoscedasticity) is provided, which is essential for the validity of any statistical tests.
- No reproducibility details are provided, which is a critical requirement for scientific rigor and independent replication.

Minor Comments:

- The manuscript does not provide a clear justification for the choice of refrigerant (R-290) based on statistical or quantitative analysis.
- The discussion of the coefficient of variation (CV) is not fully aligned with its interpretation in the context of the study.
- The use of IoT for monitoring is described in detail, but no statistical analysis of the data is provided.
- The comparison with prior work (e.g., Zavaleta, Martínez Solís) is qualitative rather than quantitative, which limits the strength of the claims.

Recommendations:

- Report confidence intervals for all key metrics such as COP, efficiency, and power consumption.
- Provide a justification for the sample size (e.g., 30 independent runs) and conduct a power analysis.
- Apply multiple comparison corrections (e.g., Bonferroni, Holm) for any statistical tests.
- Verify the assumptions of any statistical tests (e.g., normality, independence, homoscedasticity).
- Provide reproducibility details such as seeds, runs, and variance for any simulations or experiments.
- Include explicit statistical tests (e.g., Kruskal-Wallis, Wilcoxon) for any comparative claims.

FiguresTablesEquationsReviewer

Figures, tables, equations — quality and coherence with text

Score: **4.2 / 5.0** | Confidence: 95%

Strengths:

- All figures are self-explanatory and referenced in the text before they appear.
- Tables are well-structured with clear headers and consistent data with the text.
- Equations are numbered sequentially and are defined before or immediately after use.
- The manuscript provides a clear and detailed methodology that supports the claims made in the text.
- The results and discussion sections are well-aligned with the figures and tables.
- The manuscript includes a comprehensive list of references, with appropriate recency and relevance.
- The manuscript demonstrates a clear understanding of the technical and environmental implications of the proposed refrigeration system.

Weaknesses:

- The manuscript does not provide detailed descriptions of the actual experimental setup or data collection procedures, which could affect reproducibility.
- The manuscript does not include specific statistical tests or confidence intervals for the results presented, which could impact the validity of the conclusions.
- The manuscript does not provide detailed information on the calibration or validation of the IoT sensors used in the monitoring system.
- The manuscript does not provide detailed information on the experimental conditions or the exact methodology used to calculate the COP values.

- The manuscript does not provide detailed information on the experimental setup for the thermodynamic analysis, which could affect the reproducibility of the results.

Major Comments:

- The manuscript lacks detailed descriptions of the experimental setup and data collection procedures, which is critical for reproducibility in IEEE journals.
- The manuscript does not include specific statistical tests or confidence intervals for the results, which is essential for validating the claims made in the text.
- The manuscript does not provide detailed information on the calibration or validation of the IoT sensors used in the monitoring system, which is important for ensuring the accuracy of the data.

Minor Comments:

- The manuscript could benefit from more detailed descriptions of the experimental conditions and methodology used to calculate the COP values.
- The manuscript could benefit from more detailed information on the experimental setup for the thermodynamic analysis.

Recommendations:

- Include detailed descriptions of the experimental setup and data collection procedures to ensure reproducibility.
- Include specific statistical tests and confidence intervals for the results to validate the claims made in the text.
- Provide detailed information on the calibration and validation of the IoT sensors used in the monitoring system to ensure the accuracy of the data.
- Include more detailed descriptions of the experimental conditions and methodology used to calculate the COP values.
- Provide more detailed information on the experimental setup for the thermodynamic analysis.

ResultsReviewer

Results validity and empirical support

Score: **3.5 / 5.0** | Confidence: 70%

Strengths:

- The manuscript provides a comprehensive set of quantitative results including COP, heat removal rates, and energy consumption metrics.
- The results are presented in tables and figures with clear descriptions and units.
- The comparison of R-290 and R-134a is well-documented and includes relevant properties and performance metrics.
- The system design and monitoring setup are described with sufficient detail for reproducibility.

Weaknesses:

- The COP real value of 4.72 is not supported by a detailed calculation or derivation from the thermodynamic cycle data.
- The claim that the COP is 75% of the Carnot COP is not substantiated with the actual Carnot COP value under the same operating conditions.
- The results section lacks a clear statement of the statistical significance or confidence intervals for the reported performance metrics.
- The results section does not clearly state whether the COP value was calculated from experimental data or from theoretical models.
- The results section does not provide a clear explanation of how the COP was calculated from the thermodynamic states provided in Table 4.

Major Comments:

- The claim that the COP real is 4.72 and represents 75% of the Carnot COP is not supported by any detailed calculation or reference to the Carnot COP under the same conditions. This is a critical claim that requires substantiation.

- The results section lacks a clear explanation of how the COP was calculated from the thermodynamic states provided in Table 4. This is essential for reproducibility and validation.
- The manuscript does not provide a clear statement of the statistical significance or confidence intervals for the reported performance metrics. This is a major omission in the results section.

Minor Comments:

- The results section could benefit from a more detailed explanation of the thermodynamic cycle and how the COP was derived from the states in Table 4.
- The results section could include more detailed information on the experimental setup and conditions under which the COP was measured.
- The results section could benefit from a clearer distinction between theoretical and experimental results.

Recommendations:

- Provide a detailed derivation of the COP value from the thermodynamic states in Table 4.
- Include the actual Carnot COP value under the same operating conditions to substantiate the claim that the COP is 75% of the Carnot COP.
- Add statistical significance and confidence intervals for all reported performance metrics.
- Clarify whether the COP value was calculated from experimental data or from theoretical models.
- Ensure that all quantitative results are clearly tied to the corresponding figures and tables.

DiscussionConclusionsReviewer

Discussion depth and conclusions validity

Score: **2.5 / 5.0** | Confidence: 80%

Strengths:

- Clear technical calculations and equations (e.g., COPCarnot, ηII, and energy consumption estimates) are presented with numerical values and references to figures.
- The manuscript provides a direct link between the proposed system and its intended application (food preservation in rural Colombian communities).
- The discussion includes a comparison of the COP of the proposed system with other technologies (solar absorption, thermoelectric), which is relevant to the field.

Weaknesses:

- The discussion lacks a comprehensive comparison with prior work in the field, focusing only on a few references without detailed analysis.
- The manuscript does not clearly state the limitations of the study, such as assumptions made in the model, data constraints, or potential scalability issues.
- The discussion does not address unexpected findings or anomalies in the results, which is critical for a thorough analysis.
- The conclusions are vague and lack specific quantification, such as the exact impact of the system on food preservation or cost savings.
- The conclusions section is incomplete and cut off mid-sentence, making it impossible to fully assess the validity of the conclusions.

Major Comments:

- The discussion section fails to provide a thorough comparison with prior work in the field. While it mentions other technologies (solar absorption, thermoelectric), it does not analyze their performance in detail or highlight how the proposed system improves upon them.
- The manuscript does not declare any limitations of the study, which is a critical component of a scientific discussion. Limitations such as assumptions, data constraints, or model simplifications should be explicitly stated.
- The discussion does not address unexpected findings or anomalies in the results. This omission weakens the interpretation of the data and the validity of the conclusions.

• The conclusions section is incomplete and cut off, making it impossible to determine whether the conclusions are directly derived from the results or if they introduce new claims not supported by the data.

Minor Comments:

• The discussion could benefit from a more detailed explanation of the thermodynamic properties of R-290 and how they contribute to the system's performance.

• The manuscript should clarify the assumptions made in the energy consumption calculations (e.g., 40% cycle efficiency) and their impact on the results.

• The conclusion section should be completed to ensure that all claims are supported by the data and that the contribution to the field is clearly stated.

Recommendations:

• Expand the literature comparison in the discussion to include more recent and relevant studies, and provide a detailed analysis of how the proposed system differs from existing solutions.

• Clearly state the limitations of the study, such as assumptions, data constraints, or model simplifications, to provide a balanced interpretation of the results.

• Address any unexpected findings or anomalies in the results to strengthen the discussion and ensure that the conclusions are well-supported.

• Complete the conclusions section to ensure that all claims are directly derived from the results and that the contribution to the field is clearly stated.

WritingReviewer

Scientific writing quality, clarity, and style

Score: **2.5 / 5.0** | Confidence: 80%

Strengths:

• Clear technical description of the prototype and its components

• Use of relevant keywords and terminology

• Clear statement of the problem and its significance

• Use of specific data and figures to support claims

• Clear explanation of the methodology and experimental setup

Weaknesses:

• Passive voice overused, which reduces clarity and engagement

• Vague quantifiers like 'many', 'several', and 'significant' are used without numerical support

• Acronyms like DANE, DNP, FAO, IEC, ISO, and R290 are used without definition

• Paragraphs are often single sentences or run-on sentences, which reduces readability

• The abstract and introduction do not fully align with the results and discussion sections

• The methodology is not sufficiently detailed to allow for reproducibility

• The results are not fully supported by the data presented

• The discussion lacks depth and does not fully address the implications of the findings

• The conclusions are not fully supported by the data and do not clearly state the significance of the work

• The references are not fully aligned with the content and lack recent citations

Major Comments:

• The manuscript contains multiple instances of passive voice that reduce clarity and engagement. These should be revised to use active voice where possible.

• The manuscript uses vague quantifiers like 'many', 'several', and 'significant' without numerical support. These should be replaced with specific numbers to improve clarity and precision.

• The manuscript uses acronyms like DANE, DNP, FAO, IEC, ISO, and R290 without definition. These should be defined on first use to ensure clarity for all readers.

- The manuscript contains multiple paragraphs that are single sentences or run-on sentences. These should be revised to improve readability and flow.
- The abstract and introduction do not fully align with the results and discussion sections. The abstract should be revised to reflect the actual findings and conclusions of the study.
- The methodology is not sufficiently detailed to allow for reproducibility. The steps should be revised to include more specific details about the materials, equipment, and procedures used.
- The results are not fully supported by the data presented. The data should be revised to include more detailed information about the experimental conditions, measurements, and statistical analysis.
- The discussion lacks depth and does not fully address the implications of the findings. The discussion should be revised to include more detailed analysis of the results and their significance.
- The conclusions are not fully supported by the data and do not clearly state the significance of the work. The conclusions should be revised to reflect the actual findings and their implications.
- The references are not fully aligned with the content and lack recent citations. The references should be revised to include more recent and relevant sources.

Minor Comments:

- The manuscript could benefit from more consistent terminology and formatting.
- The manuscript could benefit from more precise use of technical terms.
- The manuscript could benefit from more detailed explanations of complex concepts.
- The manuscript could benefit from more specific examples to illustrate key points.
- The manuscript could benefit from more detailed descriptions of the experimental setup and procedures.

Recommendations:

- Revise the use of passive voice to improve clarity and engagement.
- Replace vague quantifiers like 'many', 'several', and 'significant' with specific numbers to improve clarity and precision.
- Define acronyms like DANE, DNP, FAO, IEC, ISO, and R290 on first use to ensure clarity for all readers.
- Revise paragraphs that are single sentences or run-on sentences to improve readability and flow.
- Revise the abstract and introduction to reflect the actual findings and conclusions of the study.
- Revise the methodology to include more specific details about the materials, equipment, and procedures used.
- Revise the results to include more detailed information about the experimental conditions, measurements, and statistical analysis.
- Revise the discussion to include more detailed analysis of the results and their significance.
- Revise the conclusions to reflect the actual findings and their implications.
- Revise the references to include more recent and relevant sources.

ReferencesReviewer

References quality, recency, and relevance

Score: **2.5 / 5.0** | Confidence: 70%

Strengths:

- References are relevant to the context of rural electrification and food preservation in Colombia.

Weaknesses:

- References lack technical depth and foundational works in thermodynamics, refrigeration cycles, and IoT monitoring.

Major Comments:

- The references section fails to include seminal works in thermodynamics, refrigeration cycles, or IoT monitoring, which are critical for establishing the technical foundation of the manuscript.

Minor Comments:

- The citation style is inconsistent, with some references formatted as URLs and others as standard citations.

Recommendations:

- Include seminal works in thermodynamics, refrigeration cycles, and IoT monitoring to strengthen the theoretical foundation of the manuscript.
- Ensure all citations follow a consistent style (e.g., IEEE or APA) and include full bibliographic details for reproducibility.

EthicsLimitationsReviewer

Research ethics, limitations, and transparency

Score: **1.0 / 5.0** | Confidence: 95%

Strengths:

- The manuscript presents a clear and structured methodology for the development of a solar-powered refrigeration system.
- The technical calculations and equations are presented with sufficient detail for a basic understanding of the system's performance.

Weaknesses:

- The manuscript lacks essential ethical reporting elements, including ethical approval, informed consent, data availability, conflict of interest, funding disclosure, and author contributions.
- Limitations are not explicitly stated, nor are they specific or honestly assessed, which is critical for a transparent and rigorous scientific report.
- Reproducibility is not addressed, as there is no mention of data or code availability, which is a key requirement for IEEE-compliant research.
- The manuscript does not provide a clear statement of the limitations of the proposed system, such as scalability, environmental impact, or long-term reliability, which are essential for a comprehensive evaluation.

Major Comments:

- The manuscript fails to meet the ethical reporting standards required for publication in IEEE journals. Ethical approval, informed consent, data availability, conflict of interest, funding disclosure, and author contributions are all missing, which is a major issue for transparency and accountability.
- The limitations section is entirely absent, which is a critical omission for a scientific manuscript. Limitations must be explicitly stated, honestly assessed, and their impact on the conclusions clearly explained.
- Reproducibility is not addressed. The manuscript does not provide information on data or code availability, which is essential for independent replication of the work.

Minor Comments:

- The methodology section is somewhat vague and lacks sufficient detail for reproducibility. For example, the exact specifications of the solar panel, refrigeration system, and measurement instruments are not provided.
- The equations are not numbered, defined, or dimensionally consistent, which is a critical requirement for IEEE-compliant research.
- The discussion section contains incomplete references and figures, which undermines the credibility and completeness of the work.

Recommendations:

- Include a statement of ethical approval for any human or animal studies, if applicable.
- Provide a data availability statement and make all data and code publicly accessible.
- Declare any potential conflicts of interest and disclose all funding sources.
- State the author contributions clearly to ensure accountability.
- Include a detailed limitations section that explicitly states the limitations of the proposed system and their impact on the conclusions.
- Ensure that all equations are numbered, defined, and dimensionally consistent.
- Provide complete references and ensure that all figures and tables are self-contained and properly labeled.

Consolidated Major Comments

These issues must be addressed before the manuscript can be accepted.

[1] [ABSTRACT] The claim of a 4.72 COP is significant, but the abstract does not specify the operating conditions or the basis for the thermodynamic analysis. This is critical for reproducibility and validation.

[2] [ABSTRACT] The phrase 'real coefficient of performance (COP) of 4.72' is not clearly defined in the abstract. The reader should be informed of the methodology used to calculate this value.

[3] The manuscript lacks a clear hypothesis or research question, which is essential for a scientific paper.

[4] The methodology is not fully described, making it impossible to assess the validity or reproducibility of the work.

[5] The results section is incomplete and lacks quantitative data to support the claims made in the abstract and conclusions.

[6] The discussion section contains an equation without proper definition, units, or context, which undermines the technical rigor.

[7] The conclusions are not supported by the results and do not reflect the actual findings of the study.

[8] The references are incomplete and contain placeholders, which is unprofessional and affects the credibility of the work.

[9] The abstract is incomplete and does not clearly state the research question or hypothesis.

[10] The 'Experiments' section is incomplete and lacks detailed experimental setup and procedure. This is a critical gap in the methodology and limits the reproducibility of the work.

[11] The 'Results' section is brief and lacks detailed statistical analysis or confidence intervals. This weakens the validity of the findings.

[12] The paper does not provide a clear description of the experimental conditions or the number of trials conducted. This is essential for assessing the reliability of the results.

Consolidated Minor Comments

These are suggestions that would improve the manuscript quality.

[1] [TITLE] The title includes a redundant English translation. It should be either in Spanish or English, not both.

[2] [KEYWORDS] The keyword 'Colombia' is specific to the study's location but may limit the global applicability of the work. Consider including broader terms such as 'off-grid communities' or 'solar-powered refrigeration'.

[3] The title is cut off and does not fully reflect the content of the manuscript.

[4] Keywords are missing and should be included to improve searchability and alignment with the content.

[5] Some equations are not consistently defined in the text, with some equations referenced without prior explanation.

[6] The paper could benefit from more detailed descriptions of the experimental conditions and the number of trials conducted.

[7] The 'Discussion' section could be more comprehensive in its analysis of the results.

[8] The manuscript does not provide a clear justification for the choice of refrigerant (R-290) based on statistical or quantitative analysis.

[9] The discussion of the coefficient of variation (CV) is not fully aligned with its interpretation in the context of the study.

[10] The use of IoT for monitoring is described in detail, but no statistical analysis of the data is provided.

[11] The comparison with prior work (e.g., Zavaleta, Martínez Solís) is qualitative rather than quantitative, which limits the strength of the claims.

[12] The manuscript could benefit from more detailed descriptions of the experimental conditions and methodology used to calculate the COP values.

Specific Improvement Recommendations

1. [TITLE] Revise the title to remove the redundant English translation and shorten it to meet the 10-20 word guideline.
2. [ABSTRACT] Clarify the thermodynamic analysis and operating conditions in the abstract to support the COP claim.
3. [ABSTRACT] Include a statement of novelty or comparison to existing solar-powered refrigeration systems in the abstract to meet IEEE standards.
4. [ABSTRACT] Define the 'real coefficient of performance (COP)' in the abstract and specify the methodology used to calculate it.
5. Complete the introduction to clearly state the research problem, hypothesis, and objectives.
6. Provide a detailed description of the methodology, including the six phases, to ensure reproducibility.
7. Include a complete experimental design and procedure in the experiments section.
8. Add quantitative data and analysis in the results section to support the claims made in the abstract and conclusions.
9. Define all equations, including units and context, in the discussion section.
10. Ensure that the conclusions are directly supported by the results and reflect the actual findings of the study.
11. Complete the references section with properly formatted citations.
12. Include keywords that accurately reflect the content of the manuscript.
